

FLOwPERlink: Improved automated snapping of streamflow permanence point observations to hydrography datasets using extensions to the hydrolink package

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Summary

Simple visual field observations on the presence or absence of surface flow in streams and other waterbodies is a straightforward and cost-effective approach for data collection that can be used for a wide range of applications related to land and water resource management. FLOW PERmanence (FLOWPER) ([Kristin L. Jaeger et al., 2020](#)), available in the Survey123® application ([ArcGIS Survey123, n.d.](#)), is one of several feature mapping applications such as Stream Tracker ([S. Kampf, 2018](#)), Crowd Water ([Seibert et al., 2019](#)), or DRYing riVER networks (DRYVER; ([Truchy et al., 2023](#))) that provides a standard protocol for the collection of surface flow presence/absence anywhere in the world. Using these data to inform resource management decisions and for inclusion in modeling exercises requires accurate linking of point observations, or snapping, to a given hydrography. In many cases, an observation taken in the field of surface water presence or absence for a given stream reach is not co-located with the line that represents that stream reach in a given hydrography. The recently revised hydrolink tool ([Wieferich et al., 2025](#)) provides automated snapping of point features to any hydrography. The FLOWPERlink package ([Pestana et al., 2025](#)) described here leverages and extends the functionality of hydrolink to prepare FLOWPER observation data for ingestion into the Probability of Streamflow Permanence (PROSPER) model ([K. L. Jaeger et al., 2019](#); [Sando et al., 2022](#)) that provides predictions on the probability of a stream reach having year-round flow.

Statement of need

Associating field observations of surface water presence/absence data to a given hydrography typically occurs manually, is not readily reproducible, can be time consuming, or requires access to proprietary software (e.g., HydroAdd; ([Demas, 2022](#))). In addition, the georeferencing procedure can be particularly uncertain or error prone in dense and complex hydrographies, such as forested or rugged landscapes, or if the horizontal accuracy of the observation's GPS coordinates was poor. The FLOWPERlink package ([Pestana et al., 2025](#)) described here leverages the hydrolink tool ([Wieferich et al., 2025](#)) for automated snapping of FLOWPER observation points and includes quality control components based on ancillary information available in the FLOWPER survey for improved snapping accuracy to a given hydrography. Specifically, FLOWPERlink uses the geographic horizontal accuracy that is part of the FLOWPER

41 observation record to better inform the snapping based on the stream name that is part of the
42 hydrolink package. Where present in the FLOWPER observation record, information about
43 the observation's location relative to tributary junctions is also used to inform the linking to
44 hydrography flowlines. In addition, information about the snapping procedure is preserved
45 in the output file for evaluation of these data in end-user applications. The FLOWPERlink
46 package can be used to georeference similar data from other crowd-sourced feature mapping
47 applications where geographic horizontal accuracy and/or tributary junction information are
48 preserved as part of the observation record.

49 Software description

50 Dependencies

51 FLOWPERlink (Pestana et al., 2025) requires Python version 3.9 or newer, and the following
52 packages: - Hydrolink $\geq$ 2.0.0 (Wieferich et al., 2025) - geopandas $\geq$ 1.0.1 (Bossche et al.,
53 2024) - sciencebasepy $\geq$ 2.0.18 (Long et al., 2023) - networkx (Hagberg et al., 2008) -
54 setuptools (The Python Packaging Authority, 2025) - requests (Reitz, 2024)

55 To leverage the tributary junction information of FLOWPER records, any hydrography dataset
56 used must include the `mainstem_flag` and `trib_jcn` fields described below. FLOWPERlink
57 includes a script to do this preprocessing with National Hydrography Dataset hydrographs.

58 Preprocessing flowlines

59 To snap FLOWPER observations to flowlines using their tributary junction information (described
60 below), the flowlines within the hydrography dataset must have the associated attributes:
61 `mainstem_flag` and `trib_jcn`. A preprocessing script was developed to derive this tributary
62 junction information from the National Hydrography Dataset (NHD; (U.S. Geological Survey,
63 2023)), NHD Plus High Resolution (NHDPlusHR; (U.S. Geological Survey, 2025b)), and NHD
64 Plus Version 2 (NHDPlusV2; (U.S. Environmental Protection Agency, 2011)) datasets. The
65 script is executable from a command line or as part of a separate Bash script and preprocesses
66 flowline data by eliminating closed loops (i.e., where flowlines representing a stream split and
67 then re-join, creating parallel flow paths), splitting the flowlines at specific distances from
68 junctions, and categorizing the flowlines as either upstream mainstem, downstream mainstem,
69 or tributary based on the network's direction information.

70 The required information is pulled from the "Flowline", "Flow", and "Value Added Attributes"
71 layers of each respective hydrography dataset. The script begins by removing divergences and
72 merging relevant fields from the "Flowline", "Flow", and "Value Added Attributes" layers with
73 the `clean_flowlines` function. The NetworkX package (Hagberg et al., 2008) is then used to
74 convert the resulting GeoPandas GeoDataFrame (Bossche et al., 2024) into a Directed Graph
75 network. As a directed graph, the code can navigate the network upstream and downstream
76 and understands how each flowline connects to the rest of the network.

77 To classify a flowline as a mainstem or a tributary, the `compute_upstream_network_length`
78 function computes the total length of all lines in the upstream network at each junction. The
79 line with the longest total upstream length is classified at this junction as the mainstem, and
80 all other lines at this junction are classified as tributaries. This allows a single stream to be
81 classified as a tributary at a downstream node where it joins a larger river but be classified as
82 a mainstem further upstream where it is joined by its own smaller tributaries. This method of
83 identifying mainstems and tributaries assumes that total line length is proportional to drainage
84 area, and that mainstems will have larger drainage areas than tributaries. This method relies
85 on adjacent basin lines having comparable detail or resolution; however, not all hydrography
86 datasets will have the same detail or resolution everywhere. For example, NHD has 1:24k lines
87 in some locations and high-resolution terrain derived 3D Hydrography Program (3DHP) lines
88 elsewhere.

To label flowlines as near a junction and as upstream or downstream of that nearby junction, the `get_junction_flowlines` and `split_flowline` functions use a `trib_jcn_dist` parameter to split all flowlines a specified distance from each junction. This creates two or three sub-flowlines from each original flowline. If the original flowline length is greater than twice the `trib_jcn_dist` value, then it will be split into three sub-flowlines, two having lengths of `trib_jcn_dist`, and the third consisting of the remainder of the flowline. The sub-flowlines are then labeled as upstream or downstream of a junction, or not near a junction, based on the flow information in the directed graph. Finally, the processed flowlines with these new fields (`mainstem_flag` and `trib_jcn`) are saved to a new file ready for hydrolinking.

However, unmapped tributary junctions, where a mainstem might be mapped in a hydrography dataset, still remains a problem. In these cases, the buffer distance parameters (described below) will control whether the point is snapped to a flowline or not snapped at all. Observations that are recorded as on a tributary, but where there is no nearby tributary to snap to (or recorded as mainstem, but where there is no nearby mainstem to snap to) are flagged with a processing message in their `trib_jcn_processing_message` field in the process described below.

Example usage

The command below illustrates usage of the Python script with an NHDPlusHR hydrography dataset, where tributary junction information is retrieved from the NHDFlowline, NHDPlusFlow, and NHDPlusFlowlineVAA layers of the geopackage file. In this example, a 30 m distance is used to define the portion of flowlines near tributary junctions (`trib_jcn_dist`), and the output is written to a new geopackage file.

```
python preprocess_flowlines.py
--flowlines_filepath NHDPLUS_H_1701_HU4_GPKG.gpkg
--flowline_layer NHDFlowline
--flowlines_identifier Permanent_Identifier
--flow_layer NHDPlusFlow
--from_identifier FromPermID
--to_identifier ToPermID
--flowlineVAA_layer NHDPlusFlowlineVAA
--flowlinesVAA_identifier NHDPlusID
--divergence_field Divergence
--trib_jcn_dist 30
--output_filepath NHDPLUS_H_1701_HU4_GPKG_preprocessed.gpkg
```

Snapping with tributary junction information

A new class, `FlowperLink`, was developed as a child class to `CustomHydrography` from `hydrolink`. This class for snapping FLOWPER, or similar point observations extends the functionality of `CustomHydrography` to consider tributary junction information in the snapping process (Figure 1). The method `trib_jcn_match` compares tributary junction information within the point dataset against tributary junction information in the flowlines dataset.

In FLOWPER records, the tributary junction type field, `TribJcnTyp`, provides context about the location of the observation relative to the stream network. The possible values for this field are 0n tributary, 0n mainstem upstream, 0n mainstem downstream, and No Data (refer to the FLOWPER User's Guide or FLOWPER Quick Guide 2.0 for more information about FLOWPER data; (Kristin L. Jaeger et al., 2020)). Based on these values, the `trib_jcn_match` method of the `FlowperLink` class applies filtering logic to possible flowline matches within the search region to narrow down the number of possible matches. If the value of `TribJcnTyp` for a FLOWPER point is No Data, all possible flowline matches remain eligible for snapping, and a processing message is recorded to indicate the lack of tributary junction information at this point. If an observation is 0n tributary then only flowlines with a corresponding tributary value in their `mainstem_flag` attribute are retained for the next steps of the

127 snapping process. If an observation is On mainstem downstream or On mainstem upstream
128 then only flowlines with mainstem_flag set as mainstem and trib_jcn set as downstream
129 of junction or upstream of junction, respectively, are retained. If none of the previous
130 conditions are met, all possible flowlines are retained, and a processing message is recorded
131 in the trib_jnc_processing_message field to indicate that matching using tributary junction
132 information was attempted but was not successful.

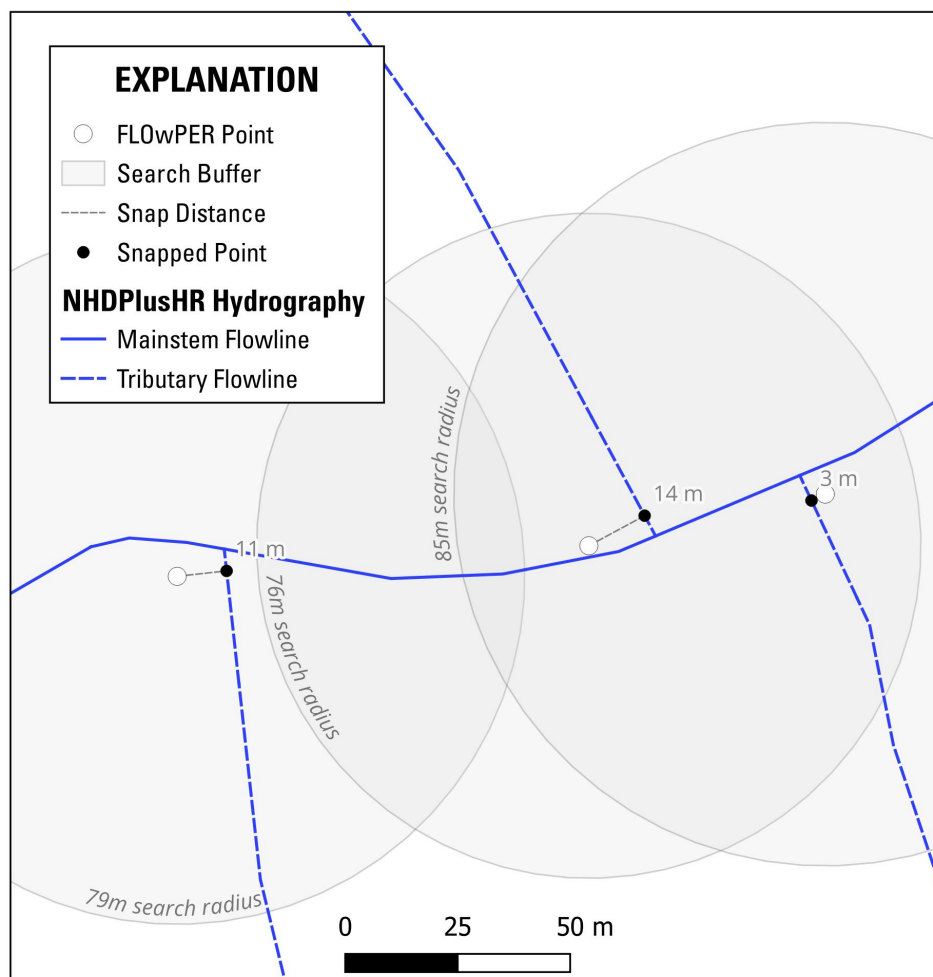


Figure 1: Map of FLOWPER observation points snapped to the NHDPlusHR hydrography dataset using tributary junction information. Though two of these FLOWPER point locations are closer to the mainstem flowline (solid line), they are snapped to the tributaries (dashed lines) based on the tributary junction information in the FLOWPER records.

133 Snapping with horizontal accuracy information

134 FLOWPER observations include a measure of the horizontal accuracy (root mean square error,
135 RMSE) of the GPS coordinates of the observation point. That information can be used to help
136 guide where on a given hydrography dataset's linework to snap the point observation [Figure 2](#).
137 The original hydrolink tool (Version 1.0.0; [Wieferich et al., 2023](#)) includes a buffer distance
138 parameter (buffer_m) to define the radius around a point of interest to search for candidate
139 lines for snapping. The revised hydrolink tool (Version 2.0.0; [Wieferich et al., 2025](#)) within
140 the CustomHydrography module can use a dynamic buffer, a different value for each point,
141 and read from a field in the point dataset (in which case buffer_m is set to the field name

rather than a static buffer distance value). Where a point might be missing a value in this field, the code relies on a `default_buffer` value which can be set to a static value or set to the mean value of the dataset.

Other parameters allow for fine-tuning this buffer distance. The `buffer_multiplier` parameter can be used to multiply the values in field `buffer_m` by a constant value. The horizontal accuracy of GPS coordinates can be impacted by the canopy cover of dense forests or surrounding mountain terrain. Therefore, using a buffer multiplier of 10 (e.g., $10/\sigma$ if the RMSE is taken to be equal to σ) is an approach that can help reduce false negatives (a point improperly identified as not belonging to any line) at the expense of increasing false positives (a point improperly identified as belonging to a line).

The RMSE, or even $10/\sigma$, may be too restrictive in cases where the RMSE is very small (e.g., $\ll 1$ m). In addition to the uncertainty in the point location, there is also uncertainty in the hydrography linework. The linework uncertainty varies based on which hydrography dataset is used due to the different spatial resolutions used to map the hydrography datasets. For example, NHD was mapped at 1:24,000 or larger scale, whereas NHDPlusV2 was mapped at 1:100,000 scale. Additionally, the PROSPER models using these observations resolve a 10-30 m spatial resolution (reach length, or grid cell size). Therefore, the linework uncertainty can be lumped with the point uncertainty by setting a lower limit on the search radius to about 10 m.

The minimum allowable buffer distance can be set based on whether stream name information (for the “name_match” method) is either present (`yes_stream_name_min_buffer`) or not present (`no_stream_name_min_buffer`). The minimum buffer distance’s dependence on stream name information allows for prioritizing the name matching method when a stream name has been recorded over the horizontal accuracy of the point’s GPS location by setting `yes_stream_name_min_buffer > no_stream_name_min_buffer`. Finally, the `max_buffer_distance` parameter sets the maximum allowable buffer distance.

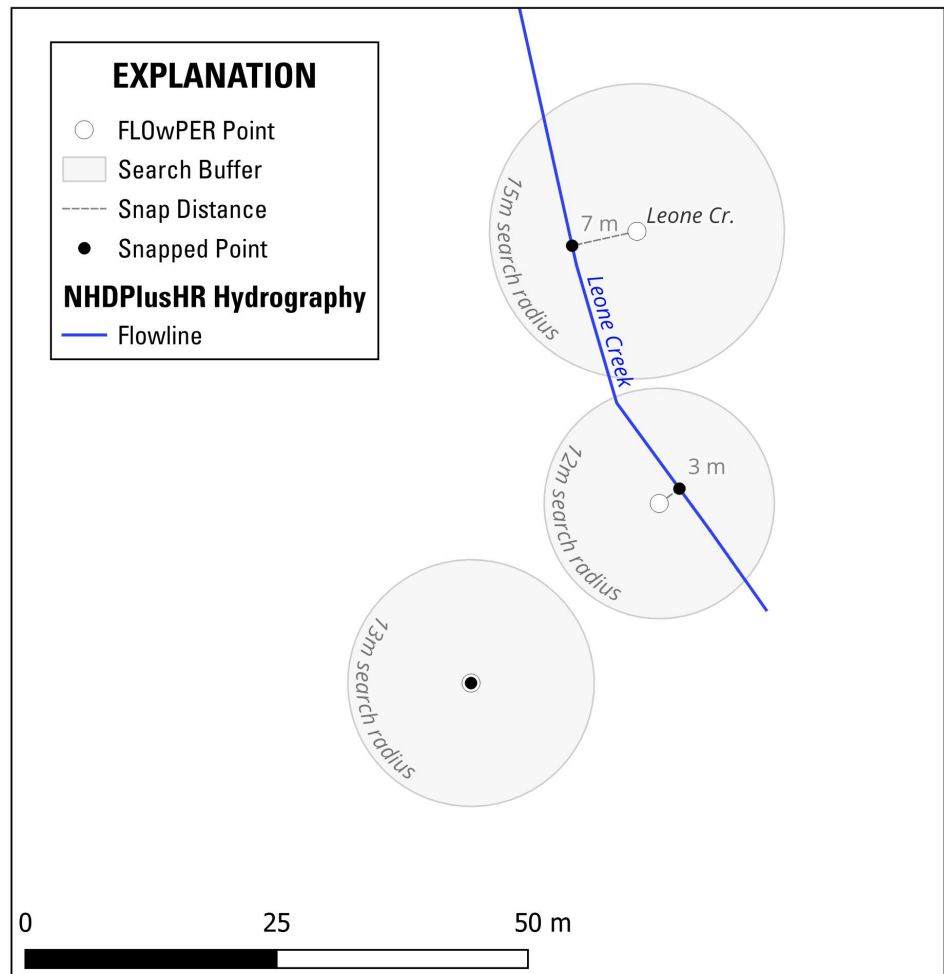


Figure 2: Map of FLOWPER observation points snapped to the NHDPlusHR hydrography dataset. The uppermost FLOWPER point snapped to a flowline based on a successful name match. The lowermost FLOWPER point did not snap to any flowline since its search buffer region did not contain any flowlines.

Example usage

The Python code below shows the use of the FlowperLink class and methods to snap FLOWPER observations to an NHDPlusHR hydrography dataset using a combination of name matching, horizontal accuracy, and tributary junction information present in the FLOWPER observation record. An instance of the FlowperLink class is initiated by passing the input datasets, specifying field names within those datasets, and setting up distance thresholds for the various snapping rules. Calling `hydrolink_method` on this object executes the snapping functions and outputs the result to a file. The last two lines in this example write out connecting lines to help visualize the snap distance and path for each input point, and the buffer region around each input point.

```
from flowperlink import FlowperLink

nhdplushr = FlowperLink(points = 'FLOWPER_points.shp',
                        flowlines = 'NHDPLUS_H_1701_HU4_GPKG_preprocessed.gpkg',
                        points_identifier = 'GlobalID',
                        flowlines_identifier = 'Permanent_Identifier',
```



```

water_name = 'Strm_Nm_Sp',
flowline_name = 'GNIS_name',
buffer_m = 'AccuracyH',
buffer_multiplier = 10,
default_buffer = 100,
no_stream_name_min_buffer = 10,
yes_stream_name_min_buffer = 15,
max_buffer_distance = 100,
keep_points_attributes = ['TribJncTyp'],
keep_flowlines_attributes = ['mainstem_flag', 'trib_jcn'])

nhdplushr.hydrolink_method(method = 'name_match',
                           hydro_type = 'flowline',
                           trib_jcn = 'TribJncTyp',
                           outfile_name = 'nhdplushr_snapped_output.gpkg',
                           similarity_cutoff = 0.6)

nhdplushr.write_connecting_lines(outfile_name='nhdplushr_snapped_connectors.gpkg')

nhdplushr.buffered_points_gdf.to_file('nhdplushr_snapped_buffer_pts.gpkg')

```

177 Snapping to TerrainWorks hydrography

178 A separate module was developed to snap FLOWPER and other streamflow permanence
 179 observations (McShane et al., 2017) to a high spatial resolution terrain derived hydrography
 180 from TerrainWorks (<https://terrainworks.com/>; (Oregon Department of Forestry (ODF) (GIS
 181 Unit and Forest Resources GIS staff), Oregon Department of Fish and Wildlife (ODFW) (Natural
 182 Resource Information Management Program) and TerrainWorks, 2024; TerrainWorks, 2014)).
 183 This hydrography necessitated its own class, TerrainWorksLink, due to the representation of
 184 flow paths not as vector lines, but as points derived from gridded data. Each point in this
 185 hydrography dataset represents the lower left coordinate of a 2-m grid cell identified as a flow
 186 path from a flow accumulation raster Figure 3. The TerrainWorks hydrography dataset does
 187 not include stream names nor the fields required to associate tributary junction information
 188 from FLOWPER observations and therefore only snaps points informed by their geographic
 189 horizontal accuracy. Methods within the TerrainWorksLink class follow the same steps as in the
 190 FlowperLink class for hydrolinking using horizontal accuracy information, albeit for the point
 191 data of the TerrainWorks hydrography. There is an additional option to offset the snapping
 192 locations from the input hydrography points. A tuple of x (Eastings) and y (Northings) values
 193 in meters (positive or negative) can be passed to the flowline_grid_offsets parameter to
 194 offset the snapping location. Options for “name_match” and “trib_jcn” methods are included
 195 for future development but will currently raise a NotImplementedError if selected.

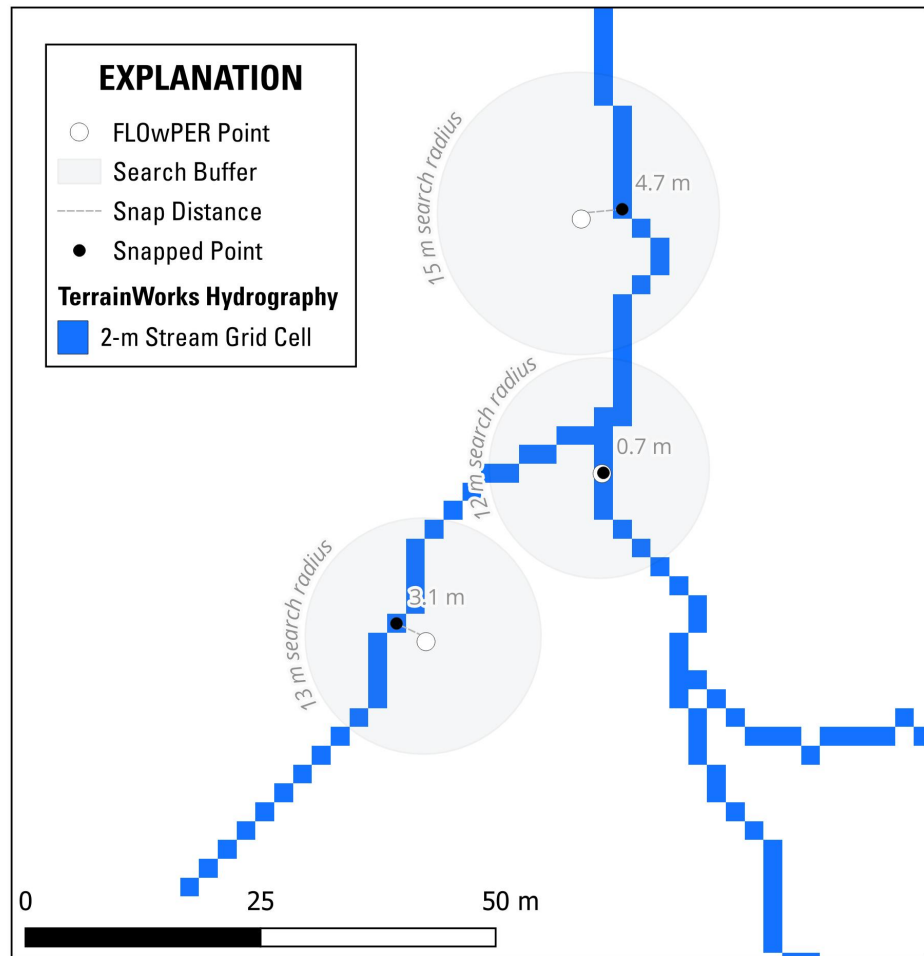


Figure 3: Map of FLOWPER observation points snapped to a 2-m gridded TerrainWorks hydrography dataset, demonstrating the dynamic search buffer distances used based on the observations' GPS horizontal accuracy.

Example usage

The Python code below shows the use of the `TerrainWorksLink` class and methods to snap FLOWPER observations to a TerrainWorks hydrography dataset using the closest distance method. Note the additional `flowline_grid_offsets` parameter. In this example, the TerrainWorks hydrography points represent the lower left corner of 2-m grid cells. To snap the FLOWPER point to grid cell centers rather than the corners, a tuple of (1, 1) is passed to this parameter to shift the snap location by 1-m east and 1-m north.

```
tw = TerrainWorksLink(points = 'FLOWPER_points.shp',
                      flowlines = 'Nodes_UpperDeschutes.gdb',
                      points_identifier='OBJECTID',
                      flowlines_identifier='NODE_ID',
                      water_name = None,
                      flowline_name = None,
                      buffer_m = 100,
                      buffer_multiplier = 1,
                      default_buffer = 100,
```



```
no_stream_name_min_buffer = 10,
yes_stream_name_min_buffer = 15,
max_buffer_distance = 100,
flowline_grid_offsets = (1, 1))
```

```
tw.hydrolink_method(method = 'closest',
trib_jcn = None,
hydro_type = 'flowline',
outfile_name = 'tw_snapped_output.gpkg',
similarity_cutoff = 0.6)
```

```
tw.write_connecting_lines(outfile_name='tw_snapped_connectors.gpkg')
tw.buffered_points_gdf.to_file('tw_snapped_buffer_pts.gpkg')
```

203 Handling FLOWPER data

204 The download_flowper.py Python script is used to download FLOWPER data from USGS
 205 ScienceBase ([U.S. Geological Survey, 2025a](#)). The script is run from the command line or
 206 through a wrapper Bash script. The script parses user input specifying the download directory
 207 and optional input comma-separated value (CSV) file path, then executes the download process
 208 based on the provided parameters. The optional CSV file input argument allows users to
 209 specify which datasets to download through a CSV file containing ScienceBase data release
 210 IDs, the unique identifiers used within ScienceBase (refer to the sciencebasepy documentation
 211 for details about interacting with ScienceBase; ([Long et al., 2023](#))). If no datasets are specified
 212 by the user, a default list of FLOWPER data releases (presented in CSV form below) from
 213 2019-2023 is downloaded instead ([Dunn & Jaeger, 2023](#); [Heaston et al., 2020, 2022, 2024](#);
 214 [Heaston, Chelgren, et al., 2025](#); [Heaston, Winter, et al., 2025](#)).

```
215 name, sb_id
216 flower_2019, 5edd08c982ce7e579c6e48db
217 flower_2020, 61b79ddddd34e78124560f8d5
218 flower_2021, 6627e820d34ea70bd5f033f9
219 flower_2022, 678fd928d34e28977994d0aa
220 flower_2023, 67e1d064d34ee7f142216699
221 rainier, 62674439d34e76103cce59d4
```

222 The load_flowper_data function reads this CSV file if provided or returns the default list. The
 223 download_flowper function then establishes a session with ScienceBase using the sciencebasepy
 224 package ([Long et al., 2023](#)), checks for the existence of a specified download directory (creating
 225 it if necessary), and iterates through the ScienceBase IDs to download relevant files (while
 226 filtering out photographs and metadata). The download_flowper function also handles the
 227 extraction of the downloaded ZIP files.

228 The handling of FLOWPER datasets is continued with the merge_flowper.py Python script.
 229 Similar to download_flowper.py, the merging script is designed to be run in the command
 230 line or through a wrapper Bash script. The script merges multiple FLOWPER shapefiles into
 231 a single Geopandas GeoDataFrame ([Bossche et al., 2024](#)) and exports the merged product
 232 to a single file (any file format supported by geopandas.to_file, e.g. shapefile, geodatabase,
 233 geopackage). All input files are reprojected into a common coordinate reference system
 234 specified by the user, or EPSG 4326 if none is provided.

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